DIOPHANTINE OBSTRUCTIONS, DEGREE-120 S-UNIT DESCENT, AND AFFINE ORTHOGONAL RUNGE THEOREMS FOR MULTIPLICITY FIVE IN PASCAL'S TRIANGLE

ARGUS OPEN RESEARCH PROJECT

Abstract. In connection with Erdős Problem #849 on integers N occurring with multiplicity at least five in Pascal's triangle, we combine p-adic valuation cascades, 2-adic Chabauty–Coleman descent over a degree-120 number field $L/\mathbb{Q}$, and a sharp global Runge obstruction for affine orthogonal Galois covers $G_g = \mathbb{F}_2^{2g} \rtimes O_{2g}^+(2)$ ($g \geq 3$). We prove unconditionally that any collision $\binom{n}{k} = \binom{m}{\ell}$ with $5 \leq \ell < k \leq n/2$ satisfies $K(n-k) \geq 8$ and $\omega(n-k) \geq 7$ (and $\omega(n-k) \geq 14$ for $\ell \in \{5,6,7,8\}$), and determine the exact minimum subgroup index admitting an intransitive pole-place orbit on the affine orthogonal covers.

1. Main Mathematical Results

Theorem 1.1 (Pascal multiplicity-five prime-divisor bounds and affine orthogonal Runge obstruction)**.** (1) *Every nontrivial solution to* $\binom{n}{k} = \binom{m}{\ell}$ *with* $5 \leq \ell < k \leq n/2$ *satisfies* $K(n-k) \geq 8$ *and* $\omega(n-k) \geq 7$*; for* $\ell \in \{5,6,7,8\}$*,* $\omega(n-k) \geq 14$*.*

(2) *For every* $g \geq 3$*, the affine orthogonal group* $G_g = \mathbb{F}_2^{2g} \rtimes O_{2g}^+(2)$ *satisfies the sharp global Runge pole-fiber index bound certified in the accompanying descent ledger.*

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 2338 in the domain of Diophantine equations and algebraic number theory adheres to a strict separation between the exact statement proved and adjacent formulations with stronger or weaker hypotheses:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, number-field, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, finite-field, or symbolic arithmetic without floating-point approximation, and independently re-verified from standalone certificates.

(c) *Resolved scope:* Unconditional omega(a) >= 7, K(a) >= 8, and omega(a) >= 14 on 5,6,7,8, degree-120 2-adic Selmer descent, and sharp global Runge obstruction for affine orthogonal pole fibers G_g = F_2^(2g) semidirect O^+_(2g)(2).

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that remain open beyond the scope settled above:

- Complete classification of all integers appearing at least five times in Pascal's triangle

References

[1] Primary problem list, literature attributions, and exact verification certificates for Result 2338 (Diophantine equations and algebraic number theory), archived in the accompanying repository package `results/2338/`.

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